DecoPro – Decompaction Workflow

Step-by-Step User Guide

1. Introduction

DecoPro is a software tool designed to calculate the decompaction of sedimentary successions and reconstruct changes in sediment thickness and porosity through time.

The software uses a porosity–depth relationship to account for the progressive reduction in sediment porosity and thickness associated with burial and compaction.

This tutorial provides a step-by-step example of how to prepare the input data, correct the depth datum, run DecoPro, and interpret decompaction calculations for an offshore well. Unlike the onshore example (user guide), offshore well depths must first be referenced to the seabed as the Datum.

2. Reference Dataset and Preparing the Input File

For this workflow, a hypothetical offshore well is used as an example. Offshore well reports commonly provide depths referenced to the top of the water column. Therefore, this workflow includes several CSV files illustrating the steps required to convert the information from an offshore well report into the final CSV input file required by DecoPro.

0_Decompaction_values_initial_table_well_report.csv: This is an optional reference table showing how well information may be presented in a well report, with layers ordered from the youngest and shallowest to the oldest and deepest. It also includes lithology and the references used for the compaction coefficient (c) and surface porosity (phi0). This file is not used as software input.

1_Decompaction_values_initial_table_well_report_seawater_datum.csv: Contains the input parameters with depths referenced to the top of the water column (sea surface). In this example, the seabed is located at a depth of 0.11 km, so z1 for the youngest sedimentary layer is 0.11 km. The layers are reordered from oldest at the top to youngest at the bottom, as required by the software. This file is not used as software input.

2_Decompaction_values_initial_table_well_report_seabed_datum.csv: Includes the original table and a second table with depths corrected to the seabed datum. The top of the youngest sedimentary layer is therefore z1 = 0 km. This file is not used as software input.

3_Offshore_well_input.csv: This is the final CSV file to be loaded into DecoPro. The file can be given any name, but it must contain the following columns in the specified order: “id, h, z2, z1, c, and phi0”.

3. Running DecoPro

Once the corrected CSV file has been prepared, the decompaction calculation can be performed using the DecoPro interface.

Follow the steps below:

1. **Launch DecoPro:** The DecoPro interface will appear, displaying an input table in the upper section and an Output panel in the lower section.
2. **Load the input file:** Click the **Load From CSV** button and select the previously prepared 3_Offshore_well_input.csv file.
3. **Run the calculation:** Once the input data have been verified, click **Submit** to initiate the calculation.
4. **View the results:** The calculated values will appear in the **Output** panel, located below the input table.

4. Understanding the Results

Once the calculation is completed, DecoPro displays the results in the Output panel, directly below the input table.

The results are organized into successive stages, beginning with the oldest layer and progressively incorporating younger overlying layers. They illustrate how burial affects layer depth, thickness, and porosity.

For each stage, DecoPro provides the following information:

Stage: The successive step in the burial and compaction calculation.

Layer base (km): The calculated depth of the base of each sedimentary layer at the corresponding stage, referenced to the seabed datum.

Average porosity: The calculated mean porosity of each layer at that stage, based on its burial depth and lithology-dependent parameters.

As younger layers are added, older layers generally become more deeply buried, and their average porosity decreases. The final stage represents the full sedimentary succession in its present-day configuration.

For this offshore example, the calculated present-day layer-base depths are referenced to the seabed datum. All calculated depths are expressed in kilometres (km), whereas porosity values are expressed as fractions between 0 and 1.

5. Applying DecoPro to New Offshore Datasets

For other offshore wells, repeat the same workflow using the appropriate water depth for each well. The datum shpudl be the seabed not the top of the water column. Ensure that the final CSV uses the seabed as zero depth and follows the required input format.